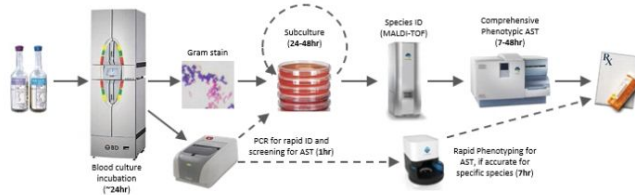


Day Zero Diagnostics (DZD)

The Future of Microbiology is Digital



Analog, culture-based diagnostics



Slow

Days to weeks to diagnose

Low Resolution

Fails to grow 40% of the time, requires multiple assays for full diagnosis

Limited

Optimized for core diagnostic use-cases



Digital, sequencing-based diagnostics



Fast

Answers within hours

High Resolution

High sensitivity, full genomic interrogation in a single diagnostic

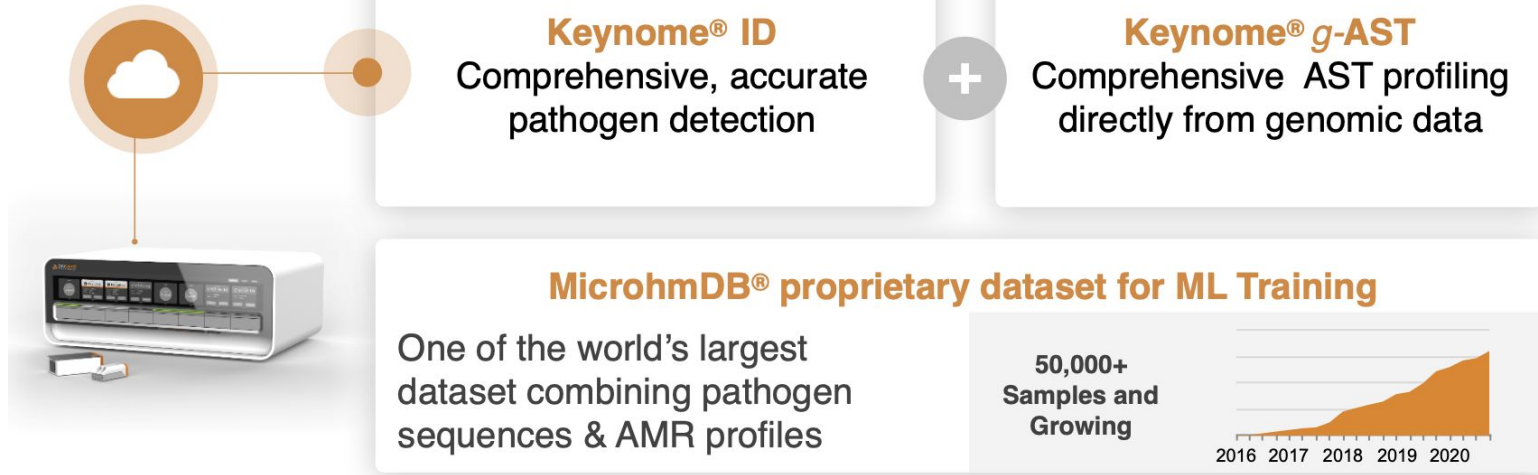
Comprehensive

Capable of comprehensive organisms, HAI & epidemiology, forecasting

Day Zero Diagnostics (DZD)

Advanced ML Algorithms

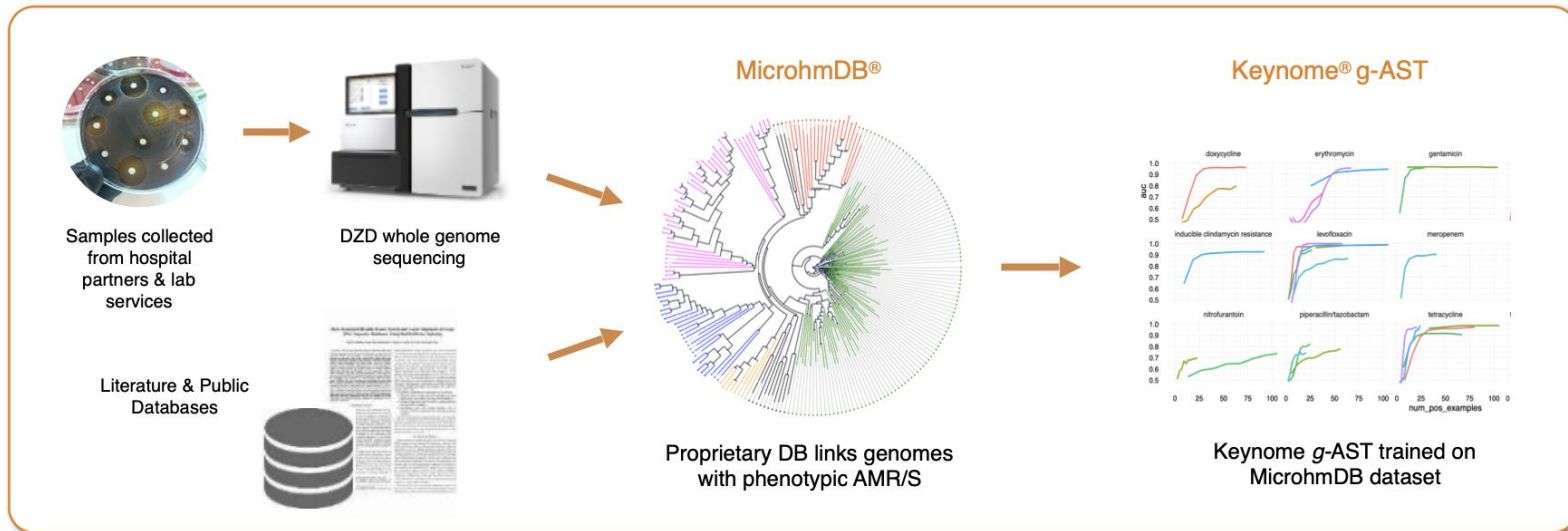
Complete diagnosis from genomic data



Day Zero Diagnostics (DZD)

Keynome® g-AST

DZD Algorithms Are Trained on MicrohmDB for AST Calls

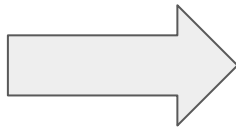


MicrohmDB: Cloud migration and versioned releases

When I took this over in late 2019, it was poorly designed and technical debt was slowing data integration and sourcing..

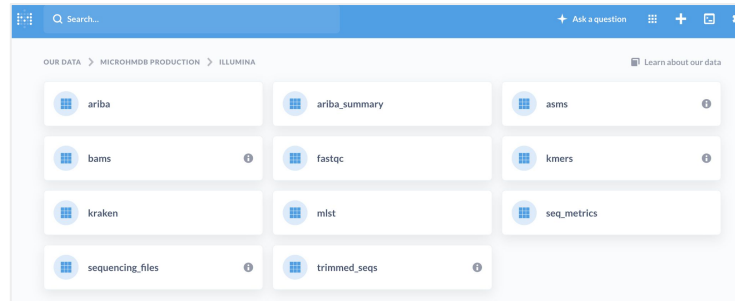
Legacy system

- Data not stored in cloud
- Compute not scalable
- Analysis not reproducible
- Data poorly modeled
- Data not versioned



MicrohmDB v1.0

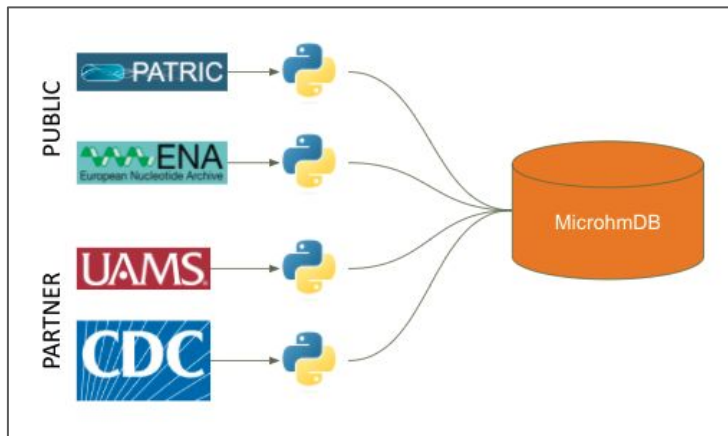
- Data stored in cloud
- Scalable compute
- Analysis reproducible
- Simper, cleaner design
- Data releases quarterly



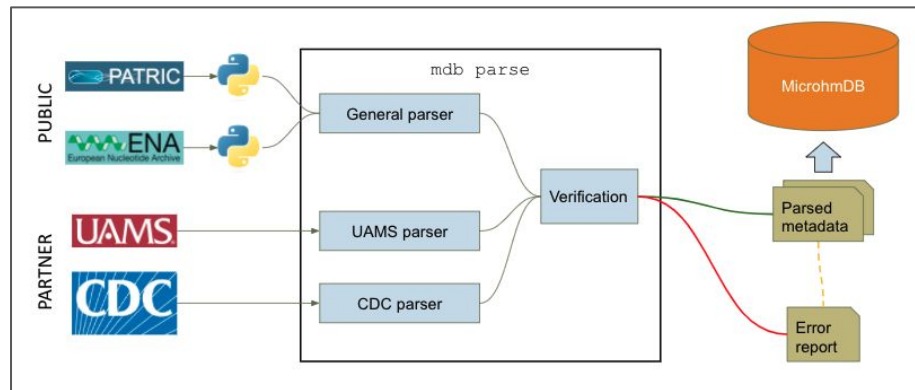
MicrohmDB: Data integration and QC

.. as we rapidly integrated new data sources, we also wanted to put data quality checks/ verification in place to ensure we were only including high quality data

Legacy system



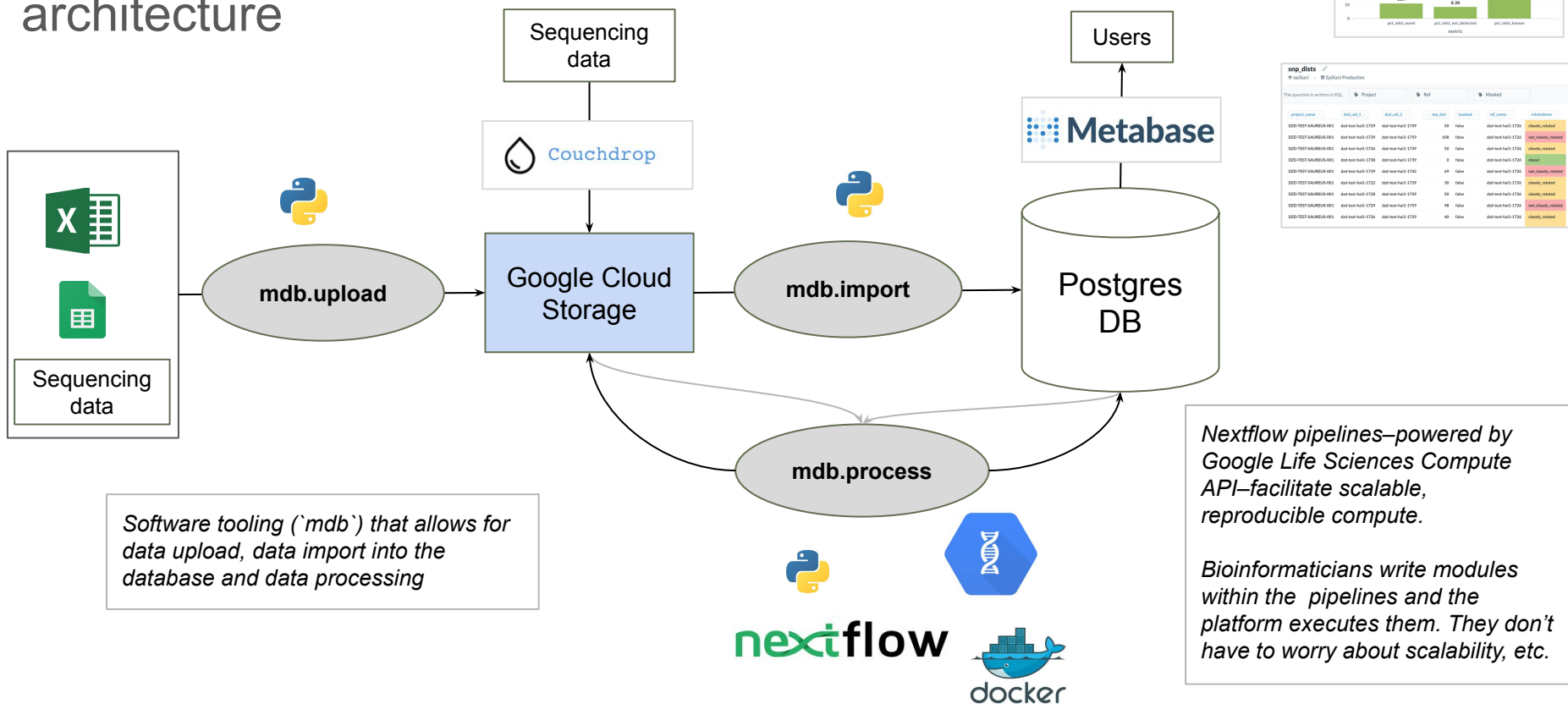
MicrohmDB v1.0



MicrohmDB: Data platform architecture

Partners and vendors can
transfer sequencing data
directly over FTP

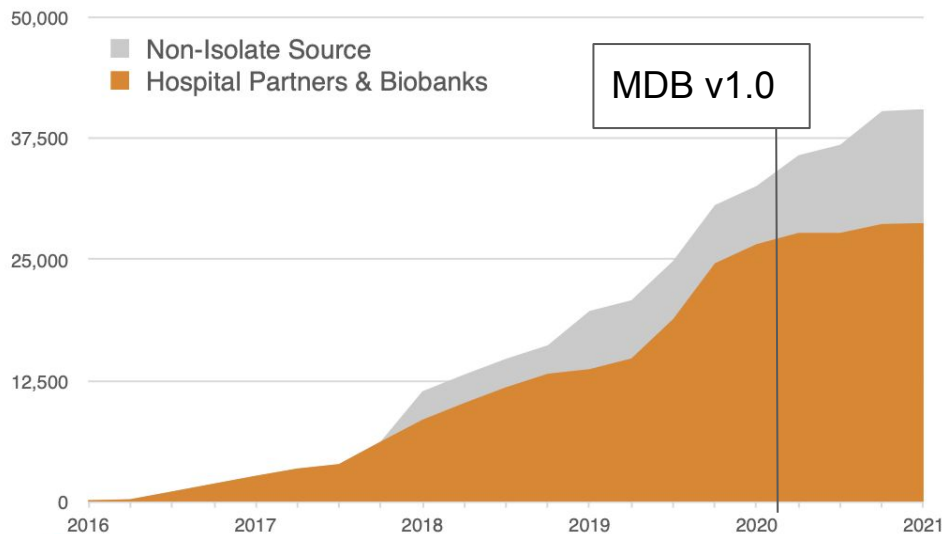
Users can interact with data
through BI dashboards, making
plots and saving common queries



MicrohmDB®

Large Scale Dataset of Pathogen Genomes and AMR Profiles

MicrohmDB: 50,000+ and growing



Customers, Contributors & Collaborators



MASSACHUSETTS
GENERAL HOSPITAL



UCIRVINE



BRIGHAM AND
WOMEN'S HOSPITAL



NEW ENGLAND BAPTIST
HOSPITAL



CANADIAN ANTIMICROBIAL
RESISTANCE ALLIANCE CARA



ARLG
Antibacterial Resistance Leadership Group

blei RESOURCES
SUPPORTING INFECTIOUS DISEASE RESEARCH

New partners since 2019